

Shambhavi Shanbhag

AI Engineer | Generative AI and Agentic Systems

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PROFESSIONAL SUMMARY

AI Engineer with nearly two years of industry and internship experience developing Generative AI, agentic AI, retrieval-augmented generation, machine learning, natural language processing, and cloud-native applications for automotive manufacturing. Built LangGraph agents, GraphRAG systems, FastAPI microservices, AWS data pipelines, and machine learning models using XGBoost and PyTorch. Experienced in LLM evaluation, human-in-the-loop workflows, Docker, MLflow, Terraform, CI/CD, and production-oriented API development, with demonstrated improvements in diagnostic speed, defect recall, pipeline cost, and process cycle time.

TECHNICAL SKILLS

Programming: Python, SQL

Generative AI and Agentic Systems: RAG, GraphRAG, LangChain, LangGraph, ReAct Agents, Tool Calling, Human-in-the-Loop Workflows, Prompt Engineering, LLM-as-Judge

Machine Learning and NLP: XGBoost, Scikit-learn, PyTorch, Hugging Face Transformers, LoRA, QLoRA, PEFT, TRL, Feature Engineering, SHAP, TF-IDF, Classification, Clustering, Model Evaluation

Retrieval, Graphs and Databases: Pinecone, Qdrant, FAISS, Neo4j, Cypher, BM25, Semantic Search, Cohere Rerank, PostgreSQL

API Development, MLOps and Testing: FastAPI, Flask, Pydantic, REST APIs, MLflow, Langfuse, RAGAS, Evidently AI, Pytest, Docker, GitHub Actions, Terraform, CI/CD

Cloud and Data Engineering: AWS Bedrock, ECS Fargate, ECR, Lambda, S3, SQS, Step Functions, Azure OpenAI, Databricks, PySpark, Delta Lake, SAP ECC

PROFESSIONAL EXPERIENCE

Mercedes-Benz India

June 2025 - June 2026

AI Engineer and Applied Data Scientist - Graduate Engineer Trainee

Pune, Maharashtra

- Agentic Production Planning:** Developed a stateful LangGraph workflow that translated annual demand into monthly production plans, applied BIS compliance rules, maintained workflow state, and routed exceptions for human approval, reducing planning cycle time by 25%.
- Manufacturing LLM Fine-Tuning:** Fine-tuned an open-source large language model on 360+ labeled maintenance scenarios using LoRA and QLoRA with PEFT and TRL to improve domain-specific maintenance reasoning.
- GraphRAG Failure-Diagnosis System:** Built an equipment-diagnosis application using Neo4j, Cypher, BM25, GraphRAG, and Streamlit to connect symptoms, components, failure modes, and maintenance evidence, reducing diagnostic information-retrieval time by approximately 40% in internal testing.
- Defect-Risk Prediction:** Trained and deployed an XGBoost classifier on 10,000+ production records using 40+ engineered features and SHAP-based feature analysis. Increased recall from 63% to 80% at 74% precision, enabling identification of 17% more defects before assembly. Tracked experiments and model versions using MLflow.
- Manufacturing Data Engineering:** Designed dimensional models and optimized Databricks SQL transformations over SAP ECC production data using common table expressions and window functions, supporting MTTR, work-in-process, scrap-rate, and operational performance dashboards.
- Warranty Failure Analytics:** Analyzed 8,000+ warranty claims and unstructured technician comments using TF-IDF, text preprocessing, and unsupervised clustering; identified 25+ recurring failure themes for engineering root-cause investigations.

Tata Motors Limited

August 2024 - April 2025

Data Science - AI and NLP Intern

Pune, Maharashtra

- RAG Optimization and Benchmarking:** Built a hybrid retrieval system combining BM25, Pinecone vector search, and Cohere reranking, improving retrieval quality for equipment troubleshooting and contributing to a 28% reduction in equipment downtime during internal evaluation.
- Enterprise Knowledge Assistant:** Developed asynchronous document-ingestion microservices using FastAPI, Docker, AWS S3, SQS, and Lambda. Implemented Pydantic validation and Pytest coverage and integrated Amazon Bedrock for production-oriented knowledge retrieval, supporting 500+ queries per month.

- **High-Throughput AWS Data Pipeline:** Architected an AWS S3 Batch Operations and Step Functions workflow that processed 10,000+ files per month, reducing compute costs by 80% and improving end-to-end processing speed by 50%.
- **Customer Complaint Intelligence:** Built a text preprocessing and supervised classification pipeline to categorize 5,000+ customer complaints across eight categories, achieving an 81% macro F1-score for high-priority escalation.

PROJECTS

Autonomous Compliance Audit Engine | *Python, FastAPI, LangGraph, Groq, Pydantic, BM25, Docker, AWS ECS Fargate, Terraform, GitHub Actions*

- Developed an agentic compliance-audit pipeline that evaluates privacy policies and financial disclosures against structured controls and returns evidence-backed, machine-readable findings.
- Implemented a ReAct audit agent with constrained tool calling, BM25 retrieval, document chunk extraction, and citation-based evidence validation to reduce unsupported conclusions.
- Designed three-run self-consistency with majority-vote consensus and strict Pydantic schemas to improve verdict stability and JSON-output reliability.
- Added an LLM-as-Judge evaluation layer to assess faithfulness, evidence alignment, completeness, and consistency across audit runs.
- Evaluated the system against 60 labeled compliance cases, achieving 86% precision, 82% recall, and 91% evidence faithfulness while limiting unsupported findings to 6% on the evaluation set.
- Containerized the FastAPI application and created Terraform and GitHub Actions workflows for deployment to AWS ECS Fargate, with an average application inference latency of approximately 15 seconds.

Repository: github.com/shambhavishanbhag1910/Legal-Compliance-Auditor-agent

Industrial Machine Anomaly Detection | *Python, PyTorch, Scikit-learn, FastAPI, Docker, Pytest, GitHub Actions, AWS ECR*

- Built and compared machine learning and autoencoder-based models for industrial equipment anomaly detection, prioritizing recall to reduce missed equipment failures.
- Developed modular preprocessing, training, evaluation, and inference components with automated Pytest coverage and GitHub Actions continuous integration.
- Containerized the inference service and published the Docker image to Amazon Elastic Container Registry for cloud deployment.

Repository: github.com/shambhavishanbhag1910/anomaly-detection

Industrial Air-Gapped RAG Research | *Python, FAISS, BM25, Local LLMs, Offline Retrieval, Manufacturing Diagnostics*

- Designed and benchmarked low-overhead offline RAG pipelines and vector database strategies for deployment on air-gapped line-side computers with no external network connectivity.
- Evaluated FAISS and BM25 retrieval with local open-source language models to reduce dependency on third-party APIs while maintaining evidence-based retrieval for manufacturing diagnostics.

EDUCATION

MKSSS's Cummins College of Engineering for Women

Master of Technology in Artificial Intelligence | CGPA: 8.9/10

Pune, Maharashtra

2023 - 2025

LEADERSHIP AND RECOGNITION

- **Technical Guest Speaker, IEEE Student Branch:** Co-conducted a technical workshop and live demonstration on database management system architecture for third-year engineering students.
- **Paint Shop Optimization Award:** Recognized at Mercedes-Benz India for manufacturing IT standardization and process improvements that contributed to more than EUR 60,000 in annual cost savings.
- **Plant-Wide Technology Enablement:** Conducted recurring training for production and IT teams on Generative AI, data engineering, and low-code process automation.